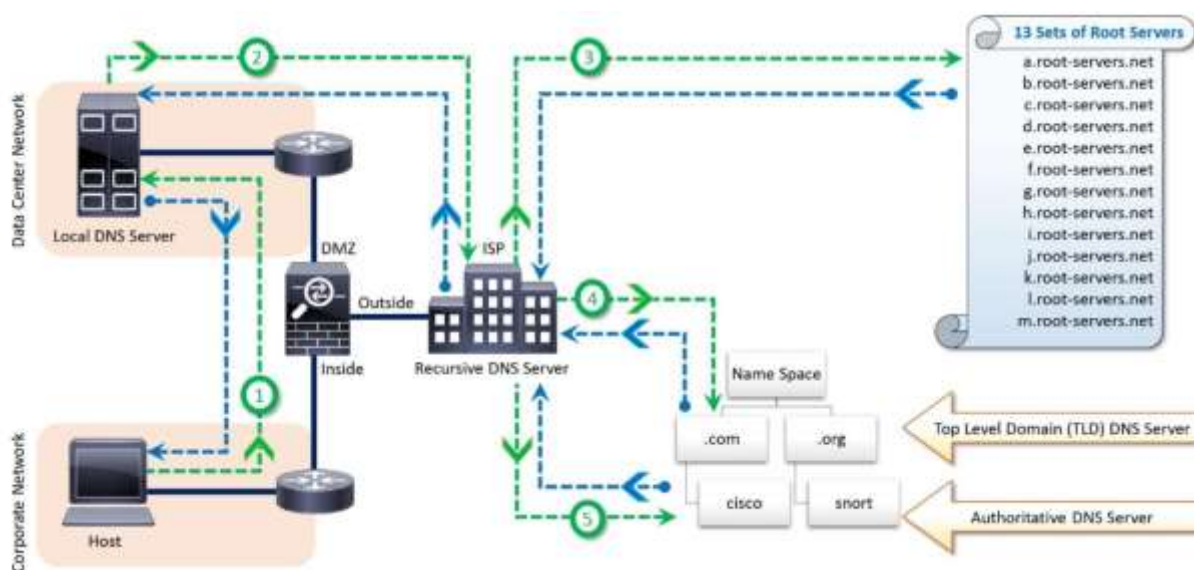
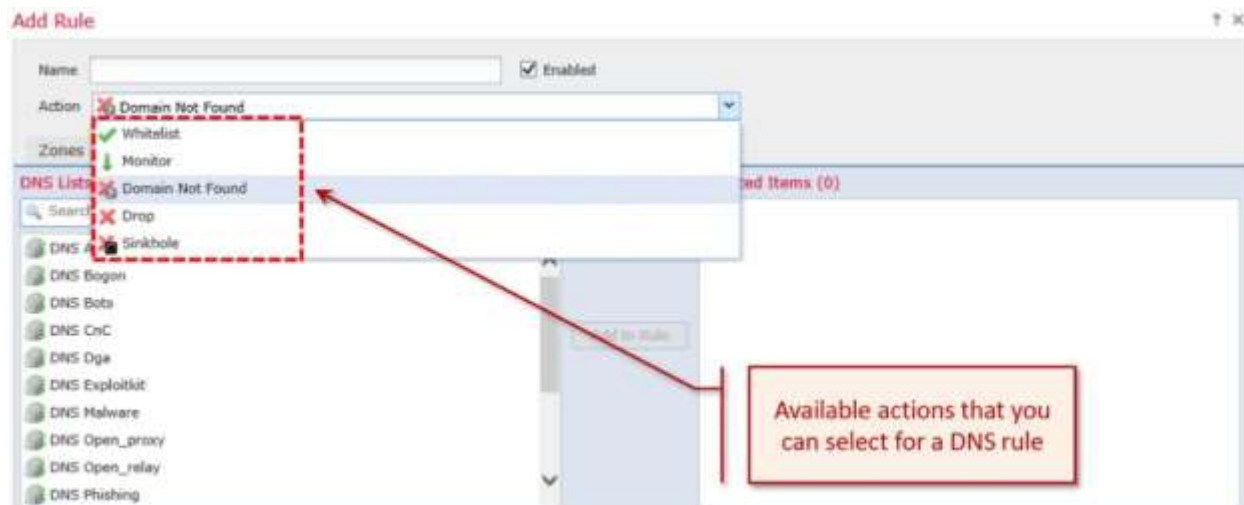


DNS Policy:

- o DNS-based Security Intelligence allows to whitelist, or blacklist traffic based on domain.
- o A DNS policy allows you to whitelist or blacklist domain name-based on connections.
- o Cisco provides the domain name (FQDN) intelligence you can use to filter your traffic.
- o Can also configure custom lists and feeds of domain names tailored to your deployment.
- o The traffic blacklisted by DNS policy is blocked and not subject to any further inspection.
- o Also, not for intrusions, exploits, malware, and so on, but also not for network discovery.
- o Firewall sits between internal & external networks; it is well placed to inspect DNS traffic.
- o Basically, DNS Polices will drop any requests for known malicious domains included list.
- o DNS policies allow to statically whitelist and blacklist FQDNs as well as dynamically Cisco.
- o DNS (Domain Name System) filtering policy can be performed in 3 three different ways.
- o TALOS maintains database of known bad DNS domains, these are updated & downloaded.
- o Filtered manually from FMC Connection Events page using Global DNS Whitelist & Blacklist.
- o The last method of Domain Name System (DNS) policy is a custom DNS Feed or custom List.
- o DNS prevent many hosts from even getting malicious IP address they are trying to connect.

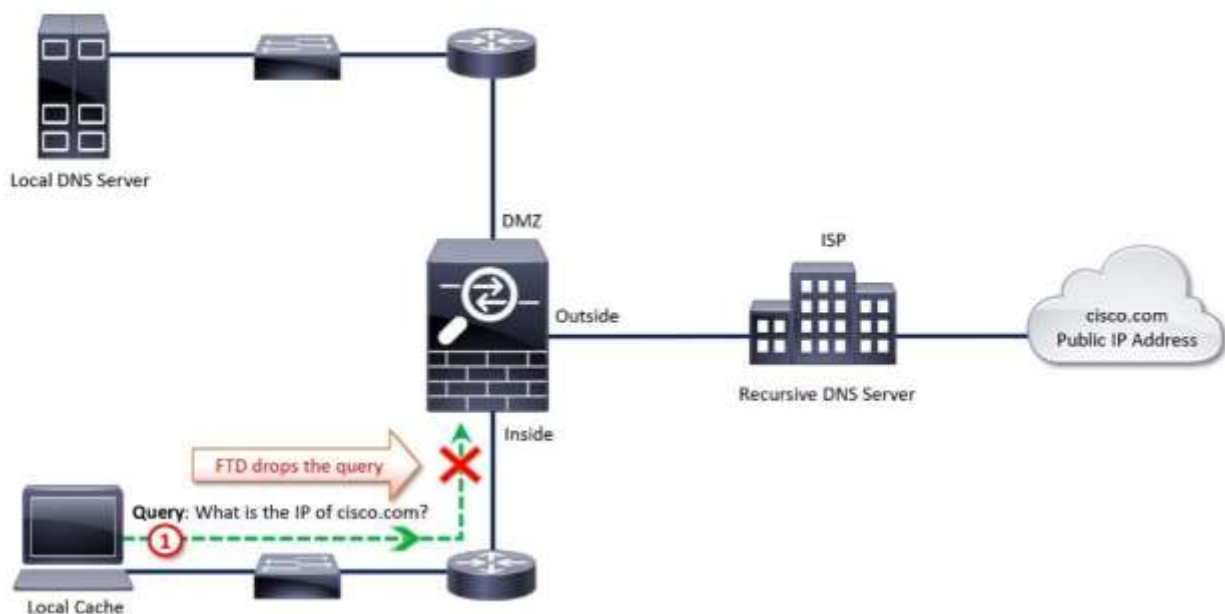


Action	Description
Whitelist	Allows matching traffic to pass.
Monitor	Does not affect traffic flow, Traffic is evaluated against other rules to determine whether it would permit or deny.
Drop (Blacklist)	Drops the traffic.
Domain Not Found	Returns a non-existent domain name response to the DNS query.
Sinkhole (Blacklist)	DNS returns a sinkhole IP address in response to the query. The sinkhole can log, or log and block.



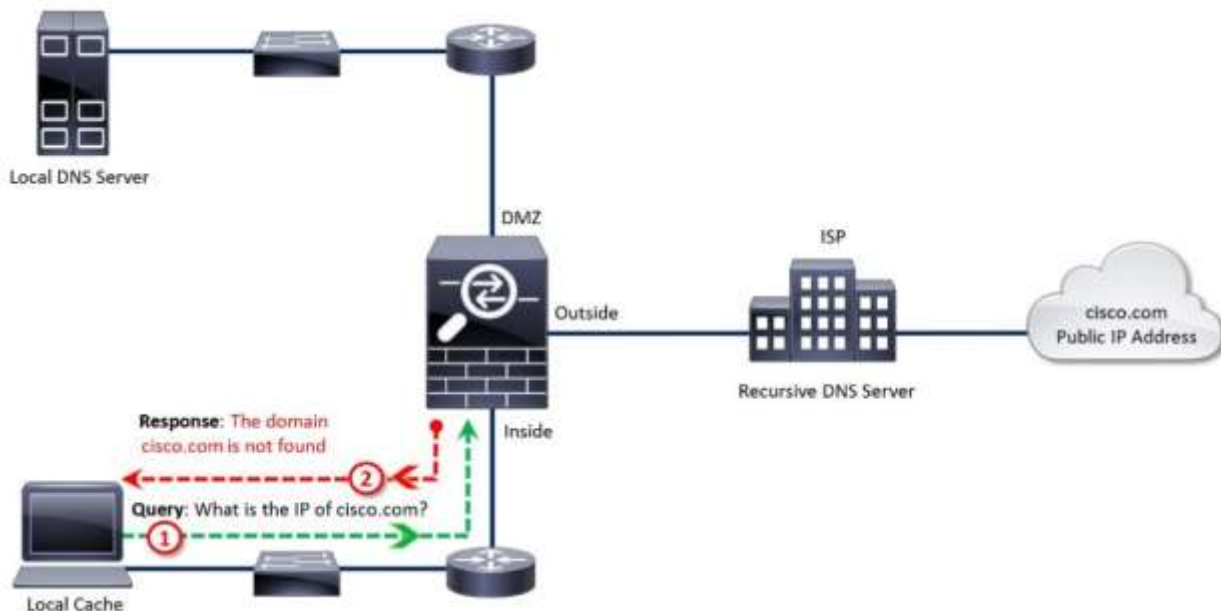
Drop Action:

FTD simply drops the DNS query for a particular domain. Drop actions are most appropriate when the Firepower device is in line between the Client and the DNS Resolver. If the Firepower device is in line between the DNS Resolver and Internet Name Servers, the Resolver (not the client) will appear to Firepower as the source of all of the requests. FTD simply drops the DNS query, no response is provided to the client.



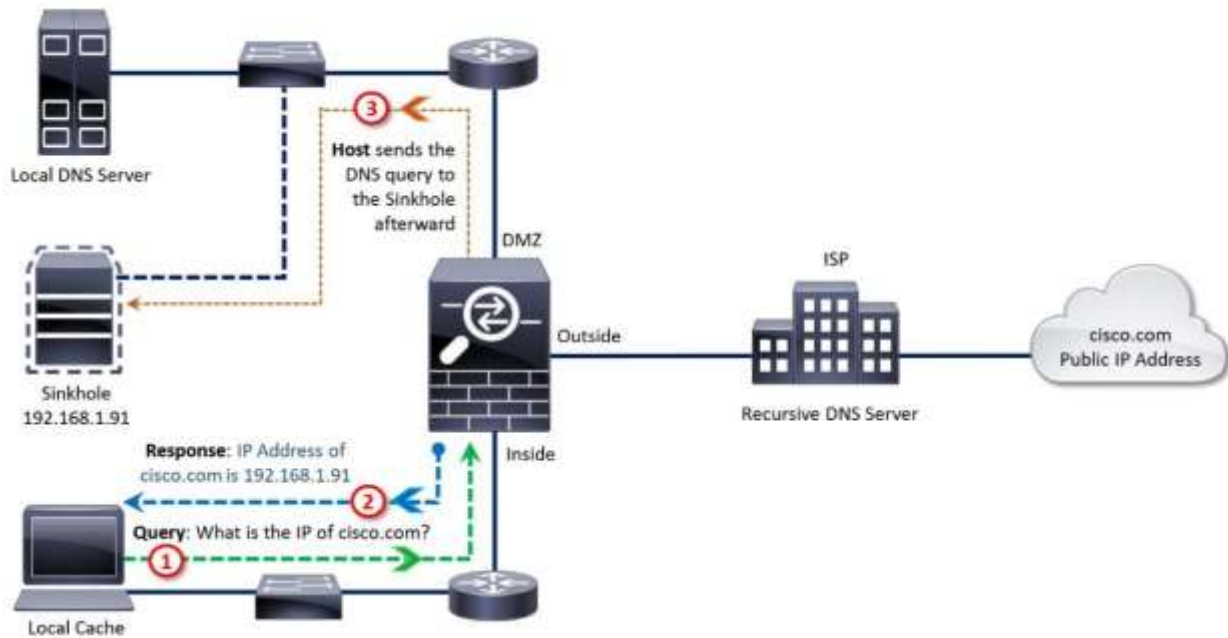
Domain Not Found Action:

The Domain Not Found action manipulates the response to contain “No such name”. This prevents the host from obtaining the destination address and discourages it from re-requesting it (as in the case of the Drop action). The Drop action works similarly but simply disallows the DNS response from reaching the client. Both the Drop and Domain Not Found actions log a Security Intelligence Event and a Start of Connection event. As a response to a DNS query, user receives NXDOMAIN (Nonexistent Domain Name) message. The NXDOMAIN message indicates that the requested domain name does not exist. Browser cannot resolve the IP address for a domain. Consequently, the user fails to access the website. Client computer that originates a DNS query receives a NXDOMAIN (Non-existent Domain Name) message as a response.

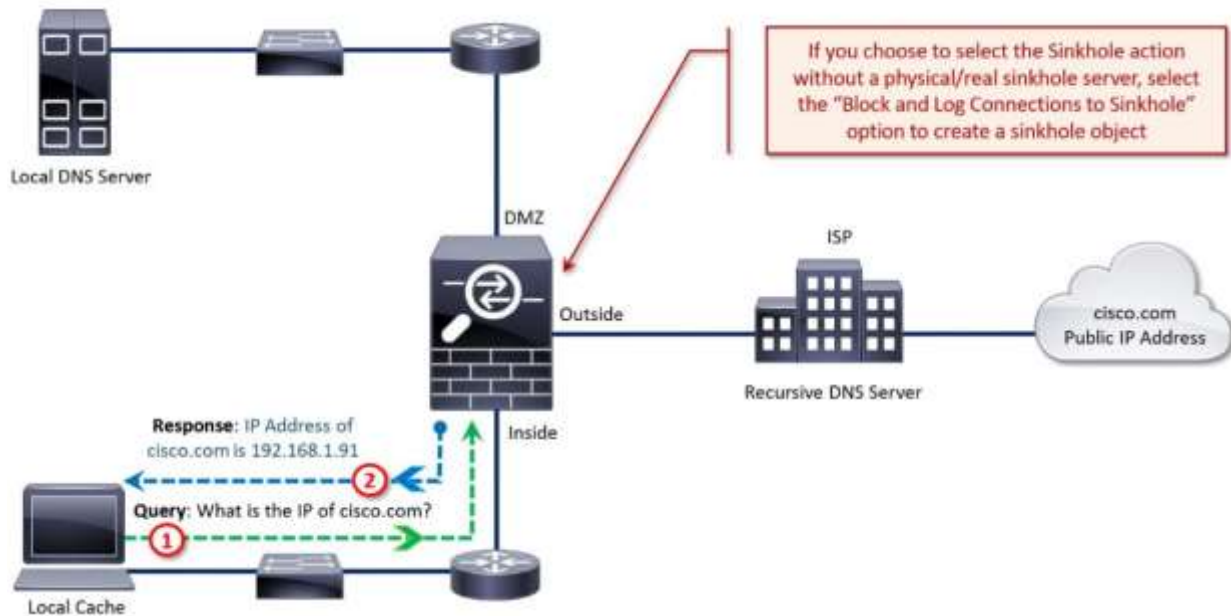


Sinkhole Action:

A DNS sinkhole is a DNS server that provides false information. Instead of returning a “No such name” DNS response to DNS queries on domains you’re blocking, it returns a fake IP address. Sinkhole changes the response to a different valid or non-valid IP address as configured by the administrator. The goal is to allow Firepower, or some system like a honeypot, to see these subsequent connection attempts. This is helpful in the case that Firepower is between the DNS Resolver and Internet Name Servers. In that case, the resolver will get the changed DNS response containing the Sinkhole IP address, cache it, and respond to the clients. The connections to the Sinkhole IP address can be observed so the misbehaving clients will be known.



Implementation of sinkhole without a physical spoof server. You can assign any false IP address within a sinkhole object. FTD uses this false address to respond to any query to a harmful domain. So, a Sinkhole with Fake Address Represents a Spoof DNS Server.



Feed:

Cisco has a dedicated threat intelligence and research team, known as Talos, who analyzes the behavior of internet traffic, performs in-depth analysis on any suspicious activities, categorizes the potential domains based on their characteristics, and list them into a file called Feeds.



CloudAgent:

One of the processes running on an FMC, CloudAgent, periodically communicates with the Cisco Cloud to download the latest feed. The frequency for updating the feed is configurable. Once an FMC downloads a feed, it sends the feed to its managed devices automatically. Redeployment of the Access Control policy is not necessary.

Nslookup:

Nslookup (**N**ame **S**erver **L**ookup) is a network administration command-line tool for querying the Domain Name System (DNS) to obtain domain name or IP address mapping, or other DNS records. You can also use the nslookup command-line tool to resolve a domain name to its IP address. The tools are available on both Windows and Linux operating systems. It allows you to view the IP address of a domain without accessing the web contents.

